9/1/2026

N + - Network Fundamentals & Building Networks.

Configuring RIP.

YAGNIK GHETIYA

TASK 1.

❖ Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.

- **Three-Router Network Topology:**

Network Setup: Three routers (Router1, Router2, and Router3) were connected using serial links in Cisco Packet Tracer.

IP Address Configuration:

Router Interface	IP Address	Subnet Mask
Router1 – Serial0/0/0	10.0.0.1	255.255.255.252
Router2 – Serial0/0/0	10.0.0.2	255.255.255.252
Router2 – Serial0/0/1	10.0.0.5	255.255.255.252
Router3 – Serial0/0/0	10.0.0.6	255.255.255.252

Result: All three routers were connected, and IP addresses were assigned to each connected interface.

TASK 2.

❖ Enable RIP version 2 on all three routers in your topology, advertise all directly connected networks, and verify that each router learns routes to all subnets using the 'show ip route' command.

- **Configure RIP Version 2 on Three Routers:**

Enable RIP v2: The following commands were configured on all three routers, with each router advertising its directly connected networks.

Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# no auto-summary
Router(config-router)# network 10.0.0.0
Router(config-router)# end

Verify Routes: The following command was used on each router:

Router# show ip route

Result: Each router learns routes to the other subnets through RIP v2. These routes appear with the code R in the routing table.

TASK 3.

❖ Change the hop count metric for one of the networks by adding an extra router between two existing routers, then observe and record how the hop count value changes in the routing tables.

- **Observing RIP Hop Count Changes:**

  Network Change: An extra router was added between two existing routers to increase the number of hops.

  Verification: The following command was used to check the routing table:

  Router# show ip route

  Observation: The hop count increased by 1 for networks reached through the additional router.

  Result: RIP updated its routing tables to reflect the new network path and hop count.

TASK 4.

❖ Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers, Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.

- **Analyzing RIP Update Packets**:

Simulation: Cisco Packet Tracer was switched to Simulation Mode, and the event filter was set to display RIP packets.

Packet Analysis:

Source	Destination	Advertised Routes
Router 1	Router 2	Directly connected and learned networks
Router 2	Router 3	Directly connected and learned networks
Router 3	Router 2	Directly connected and learned networks

Observation: RIP update packets carry routing information between routers, allowing them to update their routing tables automatically.